Software Requirements Specification

for

CampusConnect
University Club & Event Management System

Version 1.0

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Scope is intentionally limited to a manageable academic MVP.

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Revision History

Version	Date	Author	Reason for Changes
1.0	29 August 2026	Team Dooms	Initial SRS

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for **CampusConnect — University Club & Event Management System**. It describes a small web-based academic system for university clubs, events, student registration, participant management, and attendance.

The document is intended to give the project team, instructor, and testers a clear description of what the MVP should do. It deliberately excludes advanced production features so the team can complete and demonstrate the core system within a limited academic timeline.

1.2 Document Conventions

The document follows the IEEE-style SRS structure supplied for the course. Requirement identifiers such as AUTH-REQ-01 are unique within the document. **Shall** and **must** indicate required behavior; **may** indicates an optional or future capability. Priority is expressed as High, Medium, or Low.

1.3 Intended Audience and Reading Suggestions

- **Project team:** Sections 2–6 describe the scope and implementation requirements.
- **Instructor/evaluator:** Sections 1–5 provide the project purpose, users, features, and quality requirements.
- **Testers:** Section 4 provides feature-level functional requirements that can be converted into test cases.
- **Future developers:** Sections 2, 4, and 6 summarize the boundaries and planned structure.

1.4 Product Scope

CampusConnect will centralize university club and event information. Students can browse clubs and approved events, register for available events, and view their registrations. Club organizers can manage their club, create and manage events, view participants, and record attendance. Administrators can manage users and clubs, approve events, and view basic statistics.

The intended benefit is to replace scattered announcements, forms, and manual participant records with one simple platform. The system is an academic MVP, not a full university information system.

1.5 References

- Provided IEEE-style Software Requirements Specification template (Karl E. Wiegers format).
- Course: Software Engineering and System Analysis Lab, UITS.
- Project: CampusConnect — University Club & Event Management System.

2. Overall Description

2.1 Product Perspective

CampusConnect is a standalone web application. It is not intended to replace the university's complete student information system. The MVP contains a browser-based frontend, a backend application for business rules and access control, and a database for users, clubs, events, and registrations.

2.2 Product Functions

- User registration, login, logout, and role-based access.
- Club browsing and basic club management.
- Event creation, editing, deletion, and approval.
- Event browsing and detail viewing.
- Student event registration with capacity and duplicate-registration checks.
- Participant-list viewing and manual attendance recording.
- Basic administrator dashboard/statistics.

2.3 User Classes and Characteristics

User Class	Characteristics	Main Access
Student	University student; normal web user.	Browse clubs/events, register, view own registrations.
Club Organizer	Authorized club representative.	Manage club, create/manage events, view participants, record attendance.
Admin	System administrator.	Manage users/clubs, approve events, monitor basic information.

2.4 Operating Environment

- Modern web browser such as Chrome, Edge, or Firefox.
- Backend: Node.js with Express.js.
- Database: MongoDB.
- Frontend: React.js or simple HTML/CSS/JavaScript, depending on final team implementation.
- Development may use Windows or Linux, VS Code, Git/GitHub, and Postman.

2.5 Design and Implementation Constraints

- Only three roles: Student, Club Organizer, and Admin.
- Core database entities are User, Club, Event, and Registration.
- The MVP will not require AI, online payment, QR attendance, real-time chat, GPS, or a dedicated mobile application.
- Attendance will be recorded manually.
- The implementation should prioritize simple, maintainable solutions over unnecessary technologies.

2.6 User Documentation

A short user guide or presentation instructions will explain login and the main actions available to each role. Detailed production documentation is outside the MVP.

2.7 Assumptions and Dependencies

- Users have access to a modern browser and network connection when required.
- Organizers are authorized to manage their assigned club.
- Administrators approve clubs/events where approval is enabled.
- Users provide valid information.
- The team has access to the selected database and development environment.
- Email/SMS delivery is not required.

3. External Interface Requirements

3.1 User Interfaces

- Login/register pages with validation messages.
- Student interface for clubs, events, event details, registration, and personal registrations.
- Organizer interface for club information, event management, participants, and attendance.
- Admin interface for users, clubs, event approval, and basic statistics.
- Clear navigation and logout action.
- Simple error messages for invalid or incomplete input.

3.2 Hardware Interfaces

No specialized hardware is required. The system runs on standard computers/laptops and normal input devices. QR scanners, biometric devices, sensors, and other special hardware are outside the MVP.

3.3 Software Interfaces

Interface	Purpose
Web Browser	Displays the interface and communicates with the application.
Backend API	Handles authentication, clubs, events, registrations, attendance, and administration.
MongoDB	Stores users, clubs, events, and registration records.
Git/GitHub	Source-code version control and team collaboration.

3.4 Communications Interfaces

- Browser-to-server communication shall use HTTP/HTTPS.
- JSON will be used for API data where applicable.
- HTTPS should be used for a deployed system.
- No external messaging or real-time service is required for the MVP.

4. System Features

4.1 User Accounts and Authentication

4.1.1 Description and Priority

The system shall provide registration and login for the supported roles. The user's role determines accessible features.

Priority: High

4.1.2 Stimulus/Response Sequences

• User selects Register → enters required data → system validates → account is created if valid.

• User enters credentials → system validates → appropriate role dashboard is opened.

• User logs out → authenticated session/token is ended.

4.1.3 Functional Requirements

AUTH-REQ-01: The system shall allow user registration with required account information.

AUTH-REQ-02: The system shall prevent duplicate email registration.

AUTH-REQ-03: The system shall validate required registration and login fields.

AUTH-REQ-04: The system shall authenticate registered credentials.

AUTH-REQ-05: The system shall identify Student, Club Organizer, or Admin role.

AUTH-REQ-06: The system shall enforce role-based access to protected operations.

AUTH-REQ-07: The system shall provide logout.

AUTH-REQ-08: Invalid login shall produce a clear error message.

4.2 Club Management

4.2.1 Description and Priority

The system shall maintain basic university club information. Students can browse available clubs; authorized organizers/admins can manage records.

Priority: High

4.2.2 Stimulus/Response Sequences

• Organizer/Admin selects Add Club → enters details → submits → system validates and stores the club.

• Student opens Clubs → selects a club → system displays its details.

• Authorized user edits club information → system validates → updated information is stored.

4.2.3 Functional Requirements

CLUB-REQ-01: Each club shall have a unique identifier.

CLUB-REQ-02: The system shall allow an authorized user to create a club.

CLUB-REQ-03: Authorized users shall be able to update club information.

CLUB-REQ-04: Admin shall be able to manage club status.

CLUB-REQ-05: Students shall be able to view available club information.

CLUB-REQ-06: Required club fields shall be validated.

4.3 Event Management

4.3.1 Description and Priority

The system shall allow club organizers to create/manage events and admins to approve or reject pending events.

Priority: High

4.3.2 Stimulus/Response Sequences

• Organizer creates event → enters details → submits → system validates and stores it.

• Admin reviews pending event → Approve/Reject → system updates status.

• Authorized organizer edits/deletes an event → system validates the action.

4.3.3 Functional Requirements

EVENT-REQ-01: Each event shall have a unique identifier and club reference.

EVENT-REQ-02: Authorized organizers shall be able to create events.

EVENT-REQ-03: Title, date, time, location, and capacity shall be required.

EVENT-REQ-04: Organizers shall be able to edit events they manage, subject to status.

EVENT-REQ-05: Authorized organizers shall be able to delete events where permitted.

EVENT-REQ-06: Event status shall support Pending, Approved, Rejected, and Closed.

EVENT-REQ-07: Admin shall be able to approve/reject pending events.

EVENT-REQ-08: Students shall not register for rejected or closed events.

EVENT-REQ-09: Approved event details shall be visible to students.

EVENT-REQ-10: Capacity shall be a positive value.

4.4 Event Registration

4.4.1 Description and Priority

Students can reserve a place in an available event. Capacity and duplicate-registration rules shall be enforced.

Priority: High

4.4.2 Stimulus/Response Sequences

• Student opens approved event → selects Register → system checks status, capacity, and previous registration.

• If valid → registration is created → success message shown.

• If full/already registered → registration is rejected with a clear reason.

4.4.3 Functional Requirements

REG-REQ-01: Authenticated Students shall be able to register for approved events.

REG-REQ-02: Registration shall store student, event, and registration date.

REG-REQ-03: Duplicate registration for the same student/event shall be prevented.

REG-REQ-04: Registration shall stop when capacity is reached.

REG-REQ-05: Successful registration shall show confirmation.

REG-REQ-06: Students shall be able to view their registrations.

REG-REQ-07: Cancellation may be implemented if time permits; it is not required for the MVP.

REG-REQ-08: Unavailable registration shall show a clear message.

4.5 Participant and Attendance Management

4.5.1 Description and Priority

Organizers can view participants and manually record attendance. QR-based attendance is outside the MVP.

Priority: High

4.5.2 Stimulus/Response Sequences

• Organizer opens event → Participants → registered students are displayed.

• Organizer marks Present/Absent → system stores attendance.

• Organizer revisits the list → saved attendance is displayed.

4.5.3 Functional Requirements

ATT-REQ-01: Authorized organizers shall be able to view registrations for their events.

ATT-REQ-02: The participant list shall show basic identifying information.

ATT-REQ-03: Organizers shall be able to mark Present or Absent.

ATT-REQ-04: Attendance shall be stored with the registration.

ATT-REQ-05: Unauthorized users shall not change attendance.

ATT-REQ-06: Organizers shall be able to review saved attendance.

4.6 Administration and Basic Dashboard

4.6.1 Description and Priority

The admin feature provides essential system control and simple statistics only.

Priority: Medium

4.6.2 Stimulus/Response Sequences

• Admin logs in → dashboard → basic counts displayed.

• Admin opens users/clubs/events → records displayed → authorized action performed.

• Admin approves/rejects pending event → event status changes.

4.6.3 Functional Requirements

ADMIN-REQ-01: Only Admin shall access administrative functions.

ADMIN-REQ-02: Admin shall be able to view registered users.

ADMIN-REQ-03: Admin shall be able to manage club records.

ADMIN-REQ-04: Admin shall be able to approve/reject pending events.

ADMIN-REQ-05: Dashboard may display total users, clubs, events, and registrations.

ADMIN-REQ-06: Student and Organizer accounts shall not access admin-only functions.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Normal pages and API operations should respond without noticeable delay for the small number of users expected in the academic demonstration.
- Database operations should remain simple and efficient.

- The MVP is not required to support large-scale production traffic or formal load-testing targets.

5.2 Safety Requirements

The system does not control physical or safety-critical processes. No specialized safety requirements are therefore needed. Destructive actions such as deletion should use confirmation where practical.

5.3 Security Requirements

- SEC-REQ-01: Passwords shall not be stored in plain text and should use a suitable hashing mechanism.
- SEC-REQ-02: Protected API operations shall require authentication.
- SEC-REQ-03: Role-based authorization shall be enforced on the backend.
- SEC-REQ-04: User input shall be validated before storage.
- SEC-REQ-05: Users shall not modify records outside their permissions.
- SEC-REQ-06: HTTPS should be used when deployed.
- SEC-REQ-07: Passwords/authentication secrets shall not be exposed in normal responses.

5.4 Software Quality Attributes

Attribute	Requirement
Usability	Interfaces should be simple enough for first-time university users.
Correctness	Capacity, duplicate registration, roles, and attendance should behave consistently.
Maintainability	Frontend, backend, and database responsibilities should remain reasonably separated.
Reliability	Successful operations should persist data and failed operations should show clear errors.
Testability	Core functions should be testable through UI and API/database checks.
Portability	The web app should run in common modern browsers and the selected project environment.

5.5 Business Rules

- BR-01: Each user has one primary role: Student, Club Organizer, or Admin.
- BR-02: Only authorized organizers manage events belonging to their club.
- BR-03: Only approved/available events accept registrations.
- BR-04: A student may not register for the same event more than once.
- BR-05: Registration closes when capacity is reached or the event is closed.
- BR-06: Attendance is recorded only for registered participants.
- BR-07: Admin has system-level authority over users, clubs, and event approval.
- BR-08: AI, payment, QR attendance, chat, GPS, and similar advanced features are outside the MVP.

6. Other Requirements

6.1 Database Requirements

The database shall persist the core information needed by the MVP. The planned conceptual entities are User, Club, Event, and Registration. Registration connects a student to an event and stores registration/attendance information.

Entity	Core Data
User	user_id, name, email, password representation, role
Club	club_id, name, description, category, status, organizer reference
Event	event_id, club reference, title, description, date, time, location, capacity, status
Registration	registration_id, event reference, student reference, registration date, attendance

The detailed ER diagram is a separate course deliverable. Technical fields such as createdAt/updatedAt may be added during implementation if needed.

6.2 Data Validation Requirements

- Required fields shall not be empty.
- Email values shall follow a basic valid format.
- Event capacity shall be positive.
- Event date/time shall be validated sufficiently to reject obviously invalid values.
- Entity references shall point to existing records.
- Duplicate student/event registrations shall be rejected.

6.3 Future Enhancements — Not Required for MVP

Possible future extensions only: QR attendance, email/push notifications, certificates, advanced search/recommendation, mobile app, online payment, real-time chat/announcements, and advanced analytics. None is a commitment for the current submission.

Appendix A: Glossary

Term	Meaning
CampusConnect	Project name for the University Club & Event Management System.
MVP	Minimum Viable Product: the smallest working version containing the core required features.
Student	User who browses clubs/events and registers for events.
Club Organizer	Authorized user who manages a club and its events.
Admin	User with system-level management permissions.
Event	A club activity such as a workshop, seminar, competition, or cultural program.
Registration	Record showing that a student registered for an event.
Attendance	Present/Absent value recorded for a registered participant.
CRUD	Create, Read, Update, Delete operations.

Appendix B: Analysis Models

The project analysis package will contain the following supporting models:

- ER Diagram — relationships among User, Club, Event, and Registration.
- Use Case Diagram — interactions of Student, Club Organizer, and Admin.
- Activity Diagram — event registration/approval workflow.
- Sequence Diagram — registration or event creation interaction.
- This SRS — textual requirements that the models and implementation should remain consistent with.

The detailed ER diagram will be prepared separately in draw.io as requested for the course.

Appendix C: To Be Determined List

No major unresolved requirements are required for the MVP. The following small implementation decisions can be finalized by the team without changing the core scope:

- Exact frontend choice: React.js or simple HTML/CSS/JavaScript.
- Exact authentication/session implementation.
- Whether registration cancellation is implemented if time permits.
- Final visual layout of the dashboards and pages.